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Robotic Task Oriented Knowledge Graph for Human-Robot Collaboration in Disassembly

Yiwen Ding[a,b,*], Wenjun Xu[a,b], Zhihao Liu[a,b], Zude Zhou[a], Duc Truong Pham[c]

[a]*School of Information Engineering, Wuhan University of Technology, Wuhan 430070, China*
[b]*Hubei Key Laboratory of Broadband Wireless Communication and Sensor Networks (Wuhan University of Technology), Wuhan 430070, China*
[c]*Department of Mechanical Engineering, University of Birmingham, Birmingham B15 2TT, UK*

* Corresponding author. Tel.: +86-18627966706; fax: +86-27-87651800. *E-mail address:* dingyiwen@whut.edu.cn

Abstract

Traditional disassembly methods, such as manual and robotic disassembly, are no longer competent for the requirement of the complexity of the disassembly product. Therefore, the human-robot collaboration concept can be introduced to realize a novel disassembly system, towards increasing the flexibility and adaptability of them. In order to facilitate the efficient and smooth human-robot collaboration in disassembly, it is necessary to make the disassembly system more intelligent. In this paper, a robotic knowledge graph is proposed to provide an assistant for those who lack the relevant knowledge to complete the disassembly task. By natural language processing method, this paper extracts entities and relationships from the disassembly data to build a knowledge base in the form of knowledge graph. Combining graph-based knowledge representation, a prototype system is developed for human to acquire, analyze and manage the disassembly knowledge. Finally, a case study demonstrates that the proposed robotic knowledge graph has savings in terms of disassembly time, idle time and human workload, and it can be applied to assist human operator in disassembly by providing human and robots with various kinds of the needed knowledge.

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Keywords: human-robot collaboration; product disassembly; knowledge graph; knowledge base;

1. Introduction

Industrial robots play a pivotal role in today's intelligent manufacturing. Human-robot collaboration, where human workers and robots work together in the shared workspace, has received popular recognition in different fields and gradually became the new frontier in industrial robotics [1, 2]. Under the current trend of integrated automation and data exchange in intelligent manufacturing, known as industry 4.0, human-robot collaborative solutions are widely supported and applied in hybrid robotic cells. The approaches that combining human technique with modern advanced automation, integrating appropriate collaborative methods, intuitive user interfaces and specific safety mechanisms can make a novel manufacturing paradigm that meets markets' demands [3, 4]. Disassembly is indispensable in the process of equipment maintenance, recycling and reusing. Furthermore, recycling of complex products could facilitate recovery of components for reuse or remanufacturing, which is consistent with the vision of a circular economy [5]. Nowadays, the disassembly task performed by industrial robots is no longer confined to moving objects, or other repetitive operations. There are more and more complex assignments, among which human operators and robots need to complement each other's skills in a shared workspace.

In this context, human and robotic agents that are to complete disassembly tasks require a great deal of specific knowledge, therefore knowledge and the management of knowledge gradually become important. The knowledge of disassembly task usually is sophisticated and embedded in distributed data silos. Knowledge graph is a realistic way to represent and operationalize knowledge by the expressive graph data model, and supports a wider and deeper range of services including fine-grained knowledge representation,

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semantic search, multidimensional knowledge display and interpretable logical reasoning and etc. With the spreading of Artificial Intelligence, knowledge engineering, which involves integrating knowledge into computer systems for the purpose of handling complicated problems normally requiring a high level of human expertise, excels in many fields. Knowledge graph is the product of knowledge engineering supported by big data and artificial intelligence. At present, the application of knowledge graph includes question answering system, semantic analysis of big data and intelligent knowledge service. More innovative applications of knowledge graph still remain to be developed. For instance, knowledge graph can enable human to intuitively interact with the robotic system in the disassembly process. Furthermore, industrial robots are conceived as collaborative partners in future manufacturing scenarios, thus being given greater autonomy and providing proactive assistance to the human operator. Hence, a robotic knowledge graph is proposed for successful and smooth human-robot collaboration in disassembly, and equips robots with knowledge and reasoning mechanisms, as well as enable human and robots to interpret, analyze, visualize, and learn from the knowledge base.

The rest of this paper is organized as follows. Section 2 gives a brief overview of knowledge graph techniques and applications in different fields. Then, section 3 presents the description of the approach that was used to construct a robotic knowledge graph system. In section 4, combining with a case study, the proposed method is verified and analyzed by a prototype system. Finally, the paper ends with some conclusions and future perspectives.

2. Related work

Knowledge graphs have become the backbone of many intelligent information systems that demand access to organized knowledge, both domain-specific and domain-independent. The term Knowledge Graph (KG) was coined by Google in 2012, referring to their use of semantic knowledge in web search, which helps alleviate the burden of manually selecting the relevant web pages for answers [6].

As a collection of human knowledge, knowledge graph describes real world entities and their interrelations and properties, organized in a graph [6]. Recently, there are also a few universal knowledge graphs, with DBpedia and YAGO2 [7, 8] being among the most prominent ones. The most distinguished contribution of these knowledge graph applications is developing a knowledge base which contains abundant universal information acquired from unrefined facts without any added interpretation. Besides the general large-scale knowledge graphs, various industries are also establishing knowledge graphs in their own fields. Knowledge graph embedding supports many applications, such as entity prediction and relation prediction [9]. In [10], a geographic knowledge graph is proposed for data analysis, and a method of information extraction from textual geoscience data is presented. And an approach to constructing a knowledge graph for cyber security is proposed in [11].

Knowledge methodology has highlighted the increasingly significant of application value in robotic fields. An ontology-based dynamic modeling method is utilized to reflect the current state and dynamic capability of industrial robots in the disassembly process [12]. Using robotics knowledge model, an intelligent industrial kitting with more agility, flexibility is designed in [13]. In [14], a robotic system that using robotic knowledge and knowledge-based task specification is presented. Towards improving the disassemblability and maintainability of products, a knowledge-based design for disassembly approach is proposed in [15]. For the automatic disassembly process, a cognitive robotic agent with disassembly related knowledge is designed in [16]. Similarly, in [17], a knowledge-based system that contains specific knowledge and skill is developed to enhance disassembly efficiency. Another intelligent robotic system with knowledge and skills contributes to improving the ergonomics of collaborative disassembly workstations is presented in [18].

These works show that knowledge graph and knowledge engineering have made progress in different areas, but little in human-robot collaborative disassembly. It is seldom done to apply knowledge graph to a disassembly system. In this paper, a robotic knowledge graph that equips the disassembly system with more intelligence is presented, which could assist human operators to achieve the goal in the disassembly task.

3. Establishment of robotic knowledge graph system

In this section, the key technologies in building and applying knowledge graph for collaborative disassembly are discussed. And a knowledge-based disassembly assistant that exploits the structured semantic knowledge of knowledge graph is proposed. As a preliminary to knowledge acquisition and application, a domain knowledge representation model for collaborative disassembly is presented firstly.

3.1. Domain knowledge representation

The idea that people understand the objective world through the concept, which is the abstract of the objective world, and the bond that connects people's cognition of the world. Concepts, entities and their relationships that exist in the objective world can be described in a structured form by the knowledge representation in knowledge graph. The behaviors that robots are expected to exhibit become increasingly complex, which entail the use of increasingly complex knowledge, as well as the requirement for human-robot collaboration. Ontology plays a crucial role in defining the concept of the robotics disassembly domain, as a universal knowledge representation model. Based on symbolic description logic and semantic web, a hybrid knowledge representation model combined with Resource Description Framework Schema (RDFS) and Web Ontology Language (OWL) is used in our robotic knowledge graph. One of the characteristics of symbolic knowledge representation is that it is convenient to describe explicit and discrete knowledge, so the inherent interpretability of knowledge can be explained.

Because the representation of knowledge is derived from the content and context of the information, an excellent knowledge representation model means that the model requires not only the capability to acquire knowledge through the disassembly task, but also the capability to reflect the relevant context. Here, a knowledge representation for human-robot collaborative disassembly is schematically shown in Fig. 1.

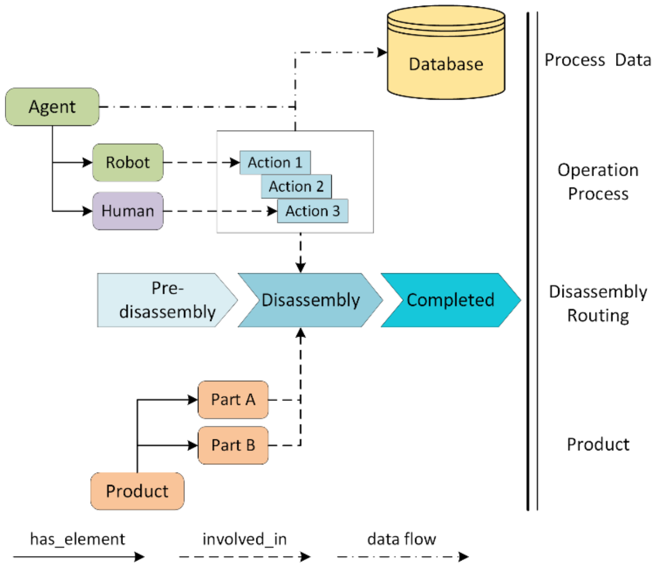

Fig. 1. Knowledge representation of a human-robot collaborative disassembly scenario.

The mentioned figure consists of a description of the *Product* that says it has two parts *Part A* and *Part B*, which can be disassembled by several actions performed by *Human* and *Robot* respectively. In the research field of knowledge management of collaborative disassembly, the knowledge of human-robot collaborative disassembly is divided into four dimensions, which consist of product dimension, disassembly routing dimension, operation process dimension and process data dimension. Product dimension contains the description of end-of-life product, such as product categories, product components and product features. The dimension of disassembly routing refers to the description of the workflow of disassembly task and detailed production schedule. In the operation process dimension, knowledge is divided into two parts according to different agent: human operator and robot. A collaborative disassembly task can be decomposed into several stages, each stage includes different operations which will be performed by a human operator, a robot and a combination of human and robot. The last dimension called process data, which mainly includes the description of disassembly record and generated data during the disassembly process, such as robotic running state and sensor's detection. The problem of creating structured knowledge out of distributed and unconnected disassembly data sources is one of the crucial challenges in knowledge graph endeavor.

3.2. Knowledge graph establishment

The robotic knowledge graph is a system that understands facts about people, robotic, activities and things and how these entities are all connected. Essentially, it is a large scale semantic network which consists of entities as well as the semantic relationships among them. Similar to a multigraph in graph theory, a knowledge graph G can be defined as a directed, node-labeled and edge-labeled multigraph, and can be expressed by a 4-tuple：

$$G = < V, E, T, P > \tag{1}$$

where V refers to the set of unique knowledge nodes, E are the edges $E \subseteq V \times V$ of the graph, T are the notations including node label and relationship type and P are the properties of nodes and relationships that consist of property key and property value.

Knowledge is composed of a set of triples in a knowledge graph, each triple can be represented as follows:

$$K = \{(h, r, t) | h, t \in V, r \in E\} \tag{2}$$

It means head node h and tail node t have a relationship of r in a triple K, where V and E refer to the set of knowledge nodes and edges respectively defined in the equation (1). It is hoped that members of a correct triple K should meet $\vec{h} + \vec{r} \approx \vec{t}$, after obtaining dense low-dimensional real value vectors of knowledge. And here is the score function to measure the accuracy of knowledge:

$$f_s(h, r, t) = -\left\| \vec{h} + \vec{r} - \vec{t} \right\|_d \tag{3}$$

In the score function, d means the Euclidean Distance of vectors $\vec{h} + \vec{r}$ and $\vec{t}$, and the score can reflect the accuracy of a triple to some extent.

Focused on specific areas, domain knowledge graph (DKG) requires certain depth and completeness, such as the proposed robotic knowledge graph, which can be applied in a variety of complex analytical applications or decision support. In the following, a method of robotic knowledge graph construction is presented，making knowledge easier for human operators to understand, access and consume.

The robotic knowledge graph architecture we proposed for a human-robot collaborative disassembly scenario is classified into four hierarchies: data hierarchy, information hierarchy, knowledge hierarchy and wisdom hierarchy, which represent the transformation of data-to-information-to-knowledge-to-wisdom from the bottom to top. The proposed robotic knowledge graph architecture is shown in Fig. 2. We will then go into more detail on the functionality of each hierarchy in the knowledge graph architecture.

Firstly, at the bottom of the architecture is the data hierarchy, which is mainly about multi-source heterogeneous data acquisition, including structured data, semi-structured data and unstructured data. The major challenge in data acquisition aspect is the transformation and integration of data in the product disassembly process. According to the knowledge representation explained in section 3.1, we need to collect the basic data of product and industrial robot, operation and process data of collaborative disassembly and detail data of disassembly task and transform them into a unified data structure for storage in a database. Traditional production data acquisition mainly relies on the manual operation mode, which is prone to human errors and inefficiencies in the process of acquisition, especially in industry big data environments. In this case, large scale automatic data acquisition tool is employed in knowledge graph construction.

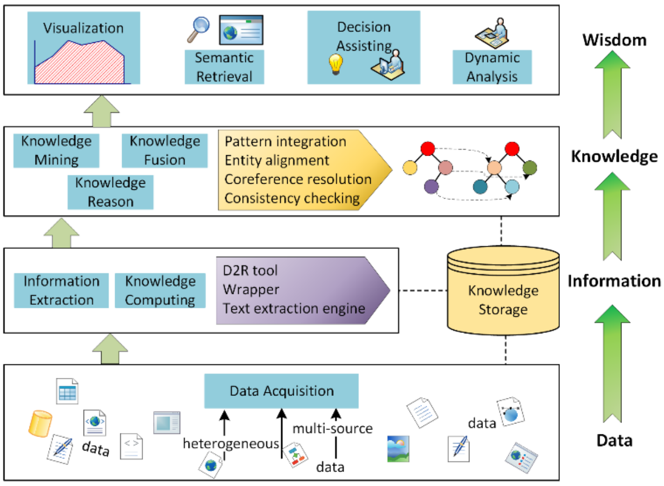

Fig. 2. Illustration of knowledge graph architecture.

The major function of the information hierarchy is to extract information from acquired raw data for knowledge computing and store extracted entities and relationships in a graph database according to the graph-based structure reflected in the equation (1). In the collaborative disassembly process, a great deal of expertise, experience and knowledge that are indispensable for human operators lacking of corresponding knowledge to work with a robot. Hence, disassembly related information must come from data and higher than data. And it is a time-sensitive, meaningful, logical, processed, and valuable data stream for decision-making. However, the traditional statistical modeling method is required to construct a large number of features manually, and there is insufficient coverage of long tail data. In this context, natural language process (NLP), semantic web and deep learning techniques can be introduced in our named entity recognition, relationship extraction and event extraction task. Compared with the traditional statistical method, the main advantage of the deep learning method is that its training is an end-to-end process, without manual definition of relevant characteristics. For different data source, associations between information of diverse modes, types and languages can be established after knowledge computing, so as to accord with task-specific knowledge representations. Furthermore, different extraction tools e.g. D2R tool, wrapper and text extraction engine are designed for multi-strategy information extraction, disassembly related domain information like robotic and product entities, disassembly action concepts and human-robot relationships can be obtained for upper knowledge hierarchy use.

Because the wide range of knowledge sources in robotic knowledge graph, there exists some problems such as the quality of knowledge is uneven, the knowledge from different data sources is duplicated, and the relationship between knowledge is not explicit enough, so it is necessary to carry out the knowledge mining and fusion task, such as coreference resolution and consistency checking. The primary function of knowledge hierarchy is knowledge processing including knowledge mining, knowledge fusion and knowledge reason. In order to facilitate the integration and retrieval of information, the quality of a knowledge graph is crucial for its applications. Knowledge fusion is a high-level knowledge organization, which enables knowledge from different knowledge sources to integrate, disambiguate, process, verify and update heterogeneous data under the unified framework, so as to achieve the fusion of data, information, methods, experience and human thoughts, and to form a high-quality knowledge base. For instance, a robotic knowledge graph ought to be consistent while updating the new disassembly task from another disassembly workstation. With the increasing of human cognitive ability, knowledge volume and service demands, knowledge graph has to be updated iteratively. Therefore, ontology is introduced in this situation, we utilize ontology matching and logical reason to make the reasoning and assessment of new knowledge, which enables knowledge graph to have a continuing reproductive capacity. On the other hand, as a DKG for specific people, the participation of experts can guarantee the validity of knowledge reasoning and fusion, and eliminate the inconsistency and redundancy of the old and new knowledge at the same time.

The wisdom hierarchy is mainly involved with using knowledge for the greater good, where operationalized knowledge has been arranged in the previous layer. As is demonstrated in Fig. 2, knowledge graphs have many applications, such as knowledge visualization, semantic retrieval, decision assistance and dynamic analysis. For intelligent robotic knowledge management service, a layer of machine comprehensibility is added to the search module by the characteristics of knowledge graph that contains richer semantic information. Thanks to the graph-based structure, knowledge graph supports traversals and queries of graph data structure and data model, respectively. So it is quite convenient for us to query, analyze and understand latent knowledge for future use. Human operators can reuse knowledge in detail such as the problems, sequence plans, operation lists and solutions of disassembly task when needed.

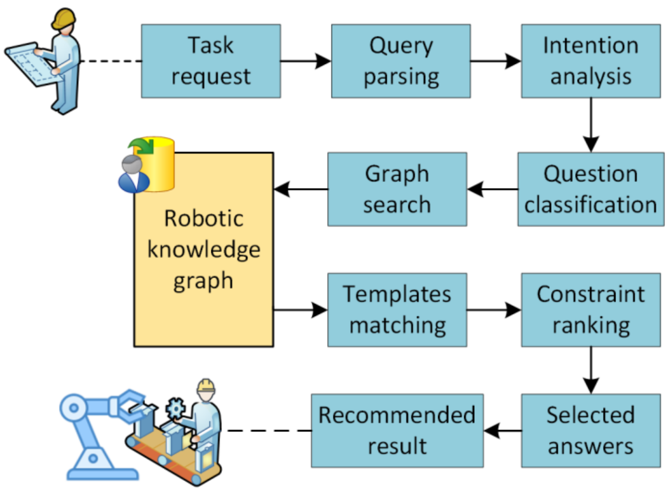

Fig. 3. Knowledge-driven disassembly service using robotic knowledge graph.

Another use of the robotic knowledge graph we build is providing effective knowledge recommendation for assisting human operator in disassembly. The proposed procedure of knowledge-driven disassembly service is illustrated in Fig. 3. During the disassembly process, a disassembly task request can be given by a human operator in order to acquire disassembly instructions. Through the analysis of the request, the problem is classified and the corresponding query statement can be generated. Incorporating graph search algorithm, then the result containing a series of candidate entities and relationships will be returned through the robotic knowledge graph. Next step is using templates to extract answers from candidate entities and relationships, with constraint ranking in the selection of answers. After answer selection, the ranking first result will be obtained. Finally the generated instructions are recommended to the human operator according to the disassembly task. And the disassembly instructions can guide the human operator and the robot to finish their own part of operations in the disassembly process. In summary, the role of wisdom hierarchy in architecture actually is to enhance the ability of knowledge graph to increase effectiveness, in order to present more application forms.

4. Case Study and Performance Analysis

In this section, a human-robot collaborative disassembly platform based on robotic knowledge graph is developed to validate the proposed method, as illustrated in Fig. 4. A collaborative robot named KUKA_iiwa is deployed, with a pneumatic gripper and industrial camera. And a disassembly product named roller chain is placed on the workbench. Besides, a robotic knowledge graph is constructed and integrated into the prototype system, which provides some benefits for process monitoring, such as providing maximum information for operational decisions or some reliable information for the disassembly process.

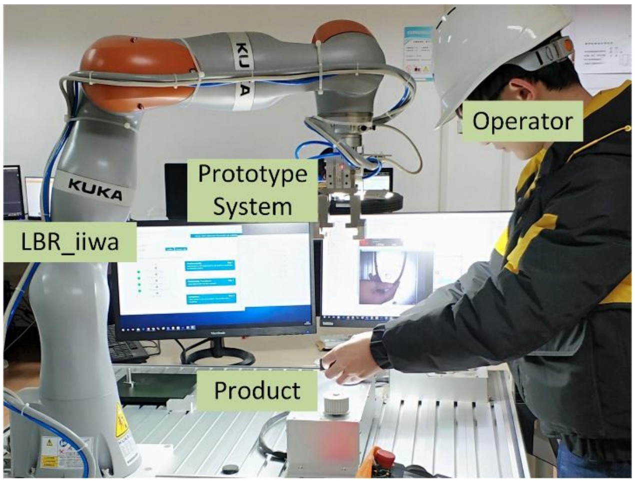

Fig. 4. Human-robot collaborative disassembly platform.

In Fig. 5, combining with an example of the knowledge management of disassembly task, an appropriate infrastructure for storing, retrieving, reusing, integrating, and analyzing disassembly data is provided. The disassembly process is observed by the prototype system, therefore human operators can apply knowledge graph to analyze what is going on and manage knowledge. Disassembly task representation has been stored in the robotic knowledge graph, which can be utilized to identify entities, concepts and attributes. Rather than just matching strings, a knowledge card that contains the supplementary information relevant to the query is presented. Combined with graph-based data visualization, such knowledge card could relieve the burden on people's side to find results manually.

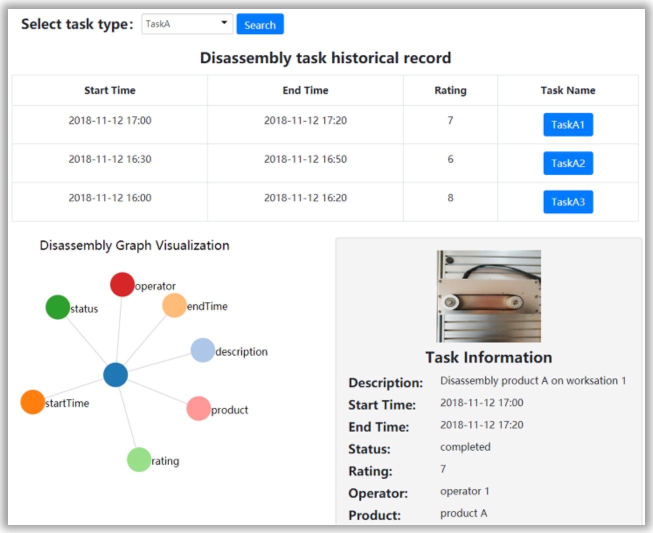

Fig. 5. Knowledge management of disassembly task.

Additionally, as illustrated in Fig. 6, a disassembly assistant based on the knowledge service (see Fig. 3) is used in the disassembly of a roller chain, which is aimed at simplifying human operator's interaction with the robot. Human operators can complete the disassembly task together with the robot according to the disassembly guide card. The disassembly steps that are performed by the robot will be converted into disassembly commands and sent to the robot execution system for collaborating with the operator. And besides the guide card, the operation execution monitoring user interface is shown, an operator can be aware of the progress of disassembly task and related information.

Furthermore, two experiments about the disassembly of roller chain are built, one of which is disassembly without knowledge graph and the other is disassembly assisted by knowledge graph. And 10 participants are involved in the experiments. Several performance indicators are adopted for the evaluation. The average of each indicator is counted, then the comparison relative indicator of two experiments is calculated for showing the degree of difference of the same indicator under different conditions. The result of the comparison between two experiments is shown in Fig. 7. After using knowledge-driven disassembly service, the idle time due to lack of relevant disassembly knowledge is less than before. And it can be observed from the result that the workload of the human is reduced. Considering the consequence acquired from experiments, the proposed robotic knowledge graph can effectively reduce product's disassembly time and improve collaborative disassembly efficiency.

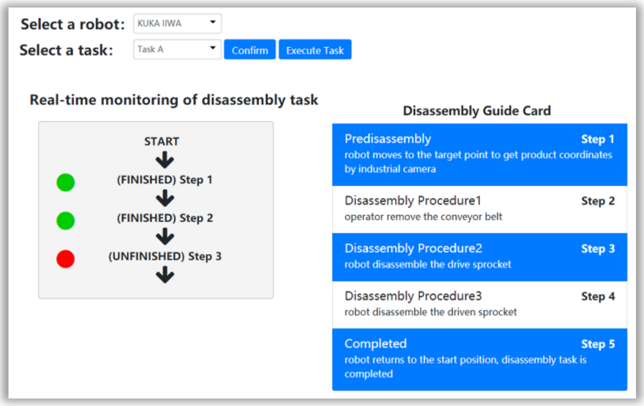

Fig. 6. Knowledge-based collaborative disassembly assistance.

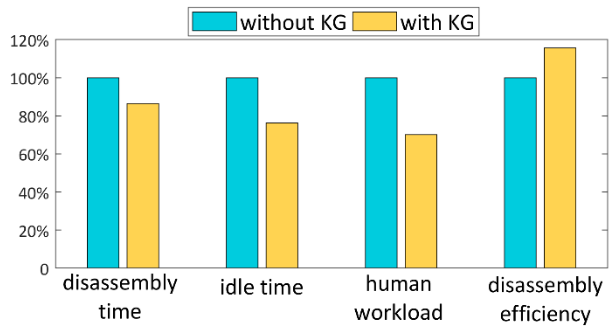

Fig. 7. Comparison of different performance indicators.

5. Conclusion and future work

Recent trends in disassembly point towards an adaptive, flexible and collaborative solutions where humans and robots share their skills. In this paper, a robotic knowledge graph for the robotic system to assist human workers to finish a disassembly task is proposed. In order to make effective use of knowledge, knowledge representation for the collaborative disassembly scenario is exhibited. Furthermore, it's shown that human operators can easily have access to all, or even most, of the relevant knowledge about disassembly task. Finally, the case study demonstrates the effectiveness of knowledge-based disassembly assistance in collaborative disassembly.

In the future, our research will address building in active bi-directional knowledge sharing between the robot and the human operator, which could enable the robot to assist human operators in a proactive way. Besides, further research involves using knowledge representation learning to improve the efficiency and accuracy of robotic knowledge graph completion will be conducted.

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References

[1] Villani V , Pini F , Leali F , et al. Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and applications. Mechatronics; 2018.

[2] Liu Z, Liu Q, Xu W, et al. Human-robot collaborative manufacturing using cooperative game: Framework and implementation. Procedia CIRP; 2018, 72: 87-92.

[3] Wang X V, Kemény Z, Váncza J, et al. Human–robot collaborative assembly in cyber-physical production: Classification framework and implementation. CIRP annals; 2017, 66(1): 5-8.

[4] Mohammed A, Wang L. Brainwaves driven human-robot collaborative assembly. Cirp Annals-manufacturing Technology; 2018, 67(1): 13-16.

[5] Liu J, Zhou Z, Pham D T, et al. Robotic disassembly sequence planning using enhanced discrete bees algorithm in remanufacturing. International Journal of Production Research; 2018, 56(9): 3134-3151.

[6] Paulheim H . Knowledge graph refinement: A survey of approaches and evaluation methods. Semantic Web; 2016, 8(3):489-508.

[7] Lehmann J , Isele R , Jakob M , et al. DBpedia – A large-scale, multilingual knowledge base extracted from Wikipedia. Semantic Web; 2015, 6(2):167-195.

[8] Hoffart J, Suchanek F M, Berberich K, et al. YAGO2: A spatially and temporally enhanced knowledge base from Wikipedia. Artificial Intelligence; 2013, 194:28-61.

[9] Jia Y, Wang Y, Jin X, et al. Path-specific knowledge graph embedding. Knowledge-Based Systems; 2018, 151: 37-44.

[10] Wang C , Ma X , Chen J , et al. Information extraction and knowledge graph construction from geoscience literature. Computers & Geosciences; 2018, 112:112-120.

[11] Jia Y, Qi Y, Shang H, et al. A practical approach to constructing a knowledge graph for cybersecurity. Engineering; 2018, 4(1): 53-60.

[12] Zheng Z, Xu W, Zhou Z, et al. Dynamic modeling of manufacturing capability for robotic disassembly in remanufacturing. Procedia Manufacturing; 2017, 10:15-25.

[13] Balakirsky S , Kootbally Z , Kramer T , et al. Knowledge driven robotics for kitting applications. Robotics and Autonomous Systems; 2013, 61(11):1205-1214.

[14] Stenmark M, Malec J. Knowledge-based instruction of manipulation tasks for industrial robotics. Robotics and Computer Integrated Manufacturing; 2015, 33:56-67.

[15] Favi C , Germani M , Mandolini M , et al. Includes knowledge of dismantling centers in the early design phase: A knowledge-based design for disassembly approach. Procedia Cirp; 2016, 48:401-406.

[16] Vongbunyong S , Kara S , Pagnucco M . Learning and revision in cognitive robotics disassembly automation. Robotics & Computer Integrated Manufacturing; 2015, 34:79-94.

[17] Lie L W, Aziz N A, Wahab D A, et al. Enhancing remanufacturing efficiency in Malaysia through a knowledge support system: A case study of brake callipers. International Journal of Industrial and Systems Engineering; 2018, 28(4): 451-467.

[18] Jungbluth J, Gerke W, Plapper P. An intelligent agent-controlled and robot-based disassembly assistant. IOP Conference Series: Materials Science and Engineering; 2017, 235(1): 012005.